

DEVOPS

- By Santhosh Kumar V S

DevOps is a software development approach where the development team and operations team work together throughout the project.

Earlier, developers focused only on coding and operations teams focused on deployment and maintenance. Because of this separation, many problems happened during software release.

DevOps was introduced to improve communication, teamwork, and software delivery speed.

DevOps mainly focuses on continuous development, continuous testing, continuous integration, continuous deployment, and continuous monitoring. Small updates are released regularly instead of very large releases. This helps companies fix errors quickly and provide better service to customers.

Real-time example:

Amazon uses DevOps practices to release website updates frequently without stopping the service. If there is any issue in payment or order tracking, the team quickly fixes it and deploys the update.

Advantages of DevOps

DevOps helps in faster software delivery because teams work together and automation reduces delays. It improves collaboration between developers and operations teams.

Continuous testing improves software quality by identifying bugs early. Automation reduces manual work and saves time. Monitoring tools help teams detect and solve issues quickly.

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Customer satisfaction also increases because updates and new features are delivered regularly.

Real-time example:

Online banking applications use DevOps to provide quick security updates and bug fixes without affecting customers.

Popular DevOps tools include Git, Jenkins, Docker, Kubernetes, and Ansible.

AGILE

Agile is a software development methodology that focuses on flexibility, teamwork, customer feedback, and continuous improvement. Instead of developing the complete software at once, Agile divides the project into smaller parts called sprints or iterations.

In traditional methods like Waterfall, changing requirements becomes difficult after development starts. Agile allows changes at any stage based on customer needs. Agile mainly focuses on delivering working software quickly and improving it continuously.

Real-time example:

A food delivery application first releases basic features like login and ordering. Later, based on customer feedback, features like live tracking and digital payments are added.

Advantages of Agile

Agile improves flexibility because changes can be added easily. Customer involvement is high throughout the project. Testing happens regularly, so product quality improves. Agile also reduces risk because issues are identified early.

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Faster delivery and better teamwork are other important advantages.

Real-time example:

E-commerce applications quickly add seasonal offers, discounts, and payment features using Agile methods.

SCRUM

Scrum is one of the most popular frameworks used in Agile. Scrum divides work into short cycles called sprints, usually lasting two to four weeks. During each sprint, the team completes selected tasks and delivers a working product.

Scrum mainly focuses on teamwork, communication, accountability, and continuous improvement.

Real-time example:

In an online learning application, one sprint may focus on login features, another sprint on video uploading, and another sprint on quizzes and certificates.

PRODUCT OWNER

The Product Owner is responsible for understanding customer requirements and managing the product backlog. They decide which features should be developed first based on business importance.

Real-time example:

In a shopping application, the Product Owner may prioritize payment options before adding extra design features.

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SCRUM MASTER

The Scrum Master helps the team follow Scrum practices correctly. They remove obstacles faced by the team and improve communication between members.

Real-time example:

If developers face server or communication problems, the Scrum Master helps solve them quickly.

ADVANTAGES OF SCRUM

Scrum improves teamwork and communication. It supports faster product delivery and continuous feedback. Problems are identified early, and product quality improves because testing and reviews happen regularly.

Real-time example:

Spotify uses Scrum teams to continuously improve music recommendations and user experience features.

DEVOPS AND AGILE TOGETHER

Agile focuses on software development and customer feedback, while DevOps focuses on deployment, automation, and maintenance. Together, they help companies develop and release software quickly and efficiently.

Real-time example:

Microsoft uses Agile teams to develop features and DevOps teams to automate deployment and monitoring for applications like Microsoft Teams.